Glaze Search, step by step: one bowl photo

Version 2026-11-06 · your photo and one typed sentence sent through the real app (today's code; colour match model 70cd01c0c673, smart search model 78afd7dfbe2c). Every number below is the app's own output, read from its answers; nothing is illustrative. One run on a test copy of the app with its own logs; recipe photos are Glazy thumbnails.

orange = from **test_similarity.zip** (the team's colour-match model: `training.ipynb`, `retrieval.ipynb`)

purple = from the **text branch** (`vendor/gc2_text_branch`: tag cleaning, cone scale, recipe descriptions)

black = the Glaze Search project **ZERO-SHOT** pretrained model used as it is

FROZEN + TRAINED HEAD pretrained model frozen + a small layer trained on our data **CLASSIC ML** classic machine learning

NO MODEL rules or formulas, no model

The two questions first

Engine	Uses 0.40 learned · 0.40 colour · 0.10 surface · 0.10 cone (4 / 4 / 1 / 1)?	Uses a latent space?	When Auto picks it
Colour match	**yes, exactly**	**yes:** their 256-d text–chemistry space	colour + cone / surface menus
Colour ranking	**their rules, learned part off:** 0 · 0.80 · 0.10 · 0.10	**no**	only a colour
Smart search	**no: its own weights (text 0.35 · photo 0.35 · formula 0.25 · colour 0.30 · cone 0.10 · surface 0.10 · …)**	**yes, its own** 256-d text–photo–formula space, trained with their method	a photo, typed words or a formula

0 · What goes in



The photo (2048 × 2048 after upload)

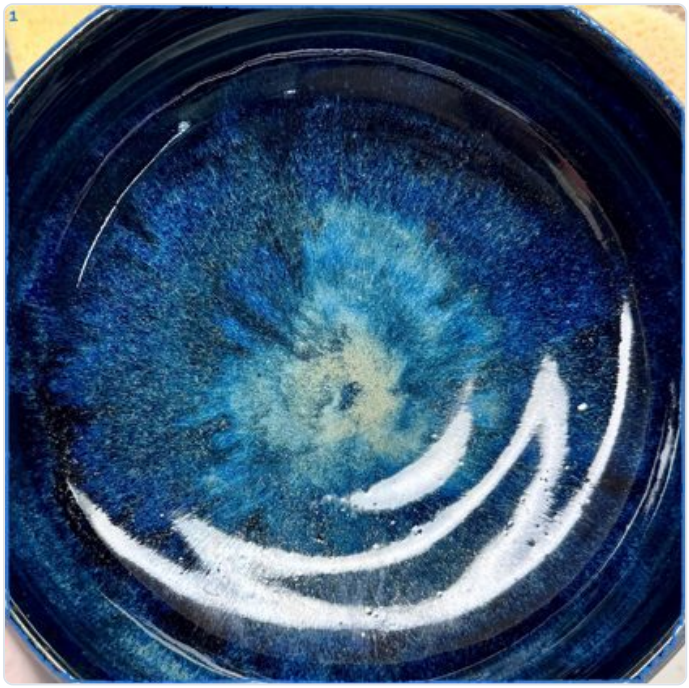
photo	a glossy dark-blue bowl seen from above, lighter blue centre
words	`shiny deep blu floating glaze with a lighter blue centre, cone 6 ox` (with typos on purpose)
formula	SiO_2 64.2 · Al_2O_3 17.8 · Fe_2O_3 1.5 (the example in the pasted text, tried in step 9)
menus	cone 6 · surface Glossy (step 7)

Part A · The photo (the same for every engine)

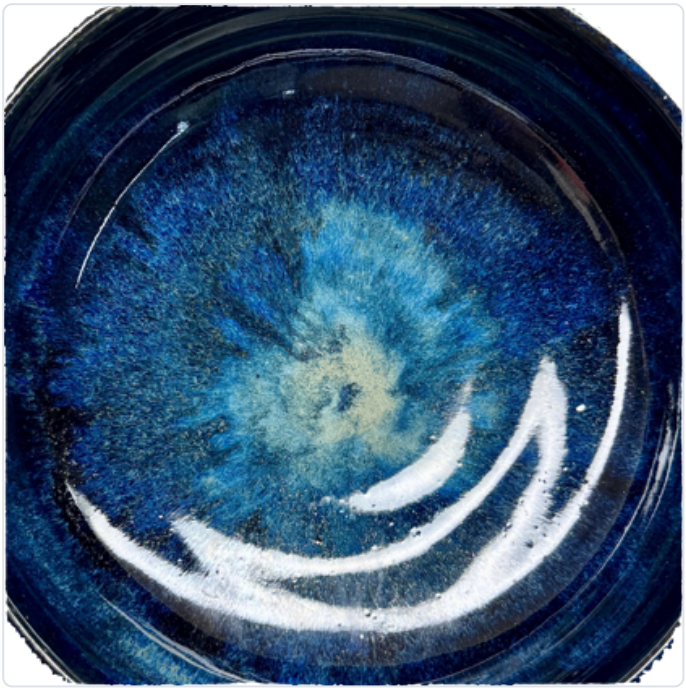
1 · Find the pot and cut it out

runs — ZERO-SHOT **Grounding DINO base** finds boxes for ceramic words · ZERO-SHOT **SAM 2.1 large** turns each box into 3 candidate outlines · ZERO-SHOT **OpenCLIP ViT-B/32** checks "is it ceramic?" and checks hand boxes before anything is cut away · rules pick the final outline

out — **1 object** filling 96 % of the photo (mode DOMINANT_OBJECT) · ceramic **1.00** · a possible hand box was rejected by CLIP · 13 s on this PC's CPU



The outline found



The cut-out object used from here on

2 · Measure the glaze colours

runs — NO MODEL **ColorThief** (median-cut palette) on the object's pixels only · k-means regions in CIELAB · each colour named by the nearest of 98 reference shades (**CIEDE2000**)

out — ■ **#0E1F3E** 56 % · ■ **#CDD8DE** 9 % · ■ **#32688D** 14 %
the light colour is mostly the window reflections · 17 colour regions (blue black, cobalt, navy, cool white …)

sent on — the main colour **#0E1F3E** ("navy blue") goes to the recipe search

3 · Six attributes (shown on "What the app found")

runs — ZERO-SHOT **Grounding DINO** asked one phrase per class ("glossy glaze . satin glaze …") → its scores → CLASSIC ML **logistic regression** trained on 1,504 labelled objects

out — shape: plate 0.92 (uncertain) — it is a bowl, but from above it looks flat · surface: glossy 0.87 · satin 0.09 (uncertain) · transparency: translucent 0.50 · opaque 0.46 (uncertain) · texture: smooth 1.00 (answered) · effect: speckled 0.72 · breaking at edges 0.23 · flowing 0.21 (uncertain) · defect: none

note — these are shown to you; the recipe search uses only what you pick in the menus or type

Part B · The words

4 · Read the sentence ("Read it")

runs NO MODEL rules + typo fixing (difflib) read the values · colour words looked up in Color Lab V3 · CLASSIC ML **TF-IDF + logistic regression** suggests tags

Your words	Read as
blu, lighter	typos fixed → blue, light
"shiny"	surface **Glossy**
"cone 6"	cone **6**
"ox"	atmosphere **Oxidation**
"deep blue"	colour **#0000FF** (Color Lab V3: generic "Blue") → model word "midnight blue"
floating, centre	no value: only smart search's text model reads these words
suggested by the TF-IDF model: surface glossy 84 % · atmosphere oxidation 78 % · transparency semi-opaque 47 %	

5 · What each engine actually receives (the reframe)

Engine	What it gets from your words	What is lost
Colour match	never your sentence: a fixed template in the training format `color: navy blue, Cone 6, Glossy, unknown, unknown` (colour word = nearest of the model's own words to the colour)	"floating", "lighter centre", "oxidation", the typos: everything outside colour · cone · surface
Smart search	the whole sentence as typed → MiniLM; the values read in step 4 (cone, surface, atmosphere) become extra score parts	nothing (the text model also reads "floating", "centre")

6 · Which engine? (Auto, decision of 2026-11-06)

runs NO MODEL the project's Auto rule (`configs/glaze_decision.json`), chosen on 395 held-out recipes

You give	Auto uses	Why (from the evaluation)
only the colour	colour ranking	With only a colour, the plain CIEDE2000 colour ranking beats both learned models: 88 % of the results in the same colour (smart search 59 %) and the recipe found more often (MRR 0.14 vs 0.11).
colour + cone / surface menus	colour match	With only the colour, cone and surface menus, colour match gives the most results with both the same colour and the same surface (63 % vs 54 % smart search); it finds the exact recipe a little less often (MRR 0.31 vs 0.34).
+ typed words, a photo object or a formula	smart search	With a photo, words or a formula, smart search finds the recipe far more often (typed sentence: in the top 10 for 79 % vs 43 % with colour match) and gives more results of the same colour and surface.

Part C · The recipe engines

7a · Only the colour → colour ranking

in colour #0E1F3E, nothing else

runs NO MODEL the colour score of retrieval.ipynb: exp(−ΔE00 / 20), its surface and cone rules, on its 6,718-recipe catalog · weights 0 · 0.80 · 0.10 · 0.10: their learned part switched off, because with only a colour it made results worse



#1 Blue Opal
Glazy #640872 · cone 6 - 8 · Satin
navy blue #162140 · ΔE00 1.6
score **0.737** = latent 0.00 · color 0.74 · surface 0.00 · cone 0.00



#2 DeBoos Recipe 100
Glazy #669264 · cone 8 - 10 · Semi-matte
blue black #0F1C36 · ΔE00 2.0
score **0.725** = latent 0.00 · color 0.73 · surface 0.00 · cone 0.00



#3 Deep blue Ice
Glazy #639978 · cone 4 - 5 · Glossy
navy blue #18223F · ΔE00 2.0
score **0.725** = latent 0.00 · color 0.72 · surface 0.00 · cone 0.00

All three are the same dark blue (ΔE00 ≈ 2), but their cones (4–10) and surfaces (Satin, Semi-matte) are mixed: nothing asked for them.

7b · Colour + cone 6 + Glossy menus → colour match (the team's model, unchanged)

in the template `color: navy blue, Cone 6, Glossy, unknown, unknown` + colour #0E1F3E

runs FROZEN + TRAINED HEAD **all-MiniLM-L6-v2** (frozen, mean pooled, 384 numbers) → text head: Linear 384 → 256 · recipe side, computed once: 26 oxides → fractions → √ → PCA-16 → chemistry head: Linear 16 → 256 · cosine between them in the shared 256-d text–chemistry latent space → learned match (cos + 1) / 2 · score 0.40 learned + 0.40 colour + 0.10 surface + 0.10 cone · Explore tab: at most 2 per K-Means cluster

```
#1 Storm Blue - Blue Room Ingredients: 0.40 × 0.773 (learned) + 0.40 × 0.751 (colour, ΔE00 5.7) + 0.10 × 1 (Glossy) + 0.10 × 1 (cone 6–8 ∋ 6) = 0.809
```



#1 Storm Blue - Blue Room Ingredients
Glazy #577829 · cone 6 - 8 · Glossy
navy blue #1F3153 · ΔE00 5.7
score **0.809** = latent 0.31 · color 0.30 · surface 0.10 · cone 0.10



#2 Cobalt blue
Glazy #561603 · cone 6 - 7 · Glossy
navy blue #262648 · ΔE00 5.7
score **0.780** = latent 0.28 · color 0.30 · surface 0.10 · cone 0.10



#3 Dolita Base - Dark Blue with black pooling
Glazy #3120 · cone 6 · Glossy
blue black #0F1932 · ΔE00 3.0
score **0.764** = latent 0.22 · color 0.34 · surface 0.10 · cone 0.10

Now every result is cone 6 and Glossy. The learned match ranks recipes whose chemistry fits the words 'navy blue, cone 6, glossy'.

8 · Words and photo → smart search (the project's model, trained with the team's method)

runs FROZEN + TRAINED HEAD **all-MiniLM-L6-v2** (frozen) → trained text head (MLP 384 → 512 → 256) · FROZEN + TRAINED HEAD **DINOv2 ViT-L/14** (frozen) on the cut-out → trained photo head (MLP 1,024 → 512 → 256) · formula: √ oxide fractions → trained head · the three heads were trained together with masked InfoNCE (temperature 0.07), recipes of the same composition group not used as wrong pairs · each recipe is one fused embedding of its text, photos and formula; its text is the colour word + the tags cleaned by the text branch (spelling, Orton cones, descriptions) · Explore tab: at most 2 per family, families by K-Means with k chosen by silhouette

score its own weights, renormalised over the parts given (not 4 / 4 / 1 / 1)

8a · + the typed sentence (parts: text 0.39 · colour 0.33 · cone 0.11 · surface 0.11 · atmosphere 0.06)



#1 Dolita Base - Dark Blue with black pooling
Glazy #3120 · cone 6 · Glossy
blue black #0F1932 · ΔE00 3.0
score **0.893** = text 0.33 · colour 0.29 · cone 0.11 · surface 0.11 · atmosphere 0.06



#2 Opal Blue
Glazy #524007 · cone 5 - 6 · Glossy
denim blue #527FC4 · ΔE00 4.4
score **0.876** = text 0.33 · colour 0.27 · cone 0.11 · surface 0.11 · atmosphere 0.06



#3 Haskap Berry (Blueberry)
Glazy #675936 · cone 6 · Glossy
cobalt blue #43577A · ΔE00 4.3
score **0.874** = text 0.33 · colour 0.27 · cone 0.11 · surface 0.11 · atmosphere 0.06

8b · + the photo object (parts: text 0.28 · photo 0.28 · colour 0.24 · cone 0.08 · surface 0.08 · atmosphere 0.04)

`#1 Studio Neves Azul GA42: 0.28 × 0.796 (text) + 0.28 × 0.729 (photo) + 0.24 × 0.860 (colour, ΔE00 3.0) + 0.08 + 0.08 + 0.04 (cone, surface, oxidation match) = 0.833`



#1 Studio Neves Azul GA42
Glazy #97643 · cone 5 1/2 - 6 · Glossy
blue black #252B3B · ΔE00 3.0
score **0.833** = text 0.22 · photo 0.20 · colour 0.21 · cone 0.08 · surface 0.08 · atmosphere 0.04



#2 Opal Blue
Glazy #794705 · cone 6 - 7 · Glossy
blue black #202538 · ΔE00 5.5
score **0.813** = text 0.22 · photo 0.21 · colour 0.18 · cone 0.08 · surface 0.08 · atmosphere 0.04



#3 Cobalt Blue
Glazy #741292 · cone 5 - 7 · Glossy
navy blue #13183B · ΔE00 4.3
score **0.811** = text 0.23 · photo 0.19 · colour 0.19 · cone 0.08 · surface 0.08 · atmosphere 0.04

8c · the photo object alone (Smart search button; parts: photo 0.54 · colour 0.46)



#1 Sunken Studio floating base - 0.16% Cobalt
Glazy #576325 · cone 7 - 8 · Glossy
blue black #0E203A · ΔE00 2.2
score **0.797** = photo 0.38 · colour 0.41



#2 Studio Neves Azul GA42
Glazy #97643 · cone 5 1/2 - 6 · Glossy
blue black #252B3B · ΔE00 3.0
score **0.790** = photo 0.39 · colour 0.40



#3 MIDDLE OF THE NIGHT
Glazy #690405 · cone 5 - 7 · Semi-glossy
blue black #121D34 · ΔE00 3.0
score **0.782** = photo 0.38 · colour 0.40

From the look alone it finds a floating blue base: the photo part matches recipes whose photos look like your bowl.

Part D · Two things to know

9 · The formula from the pasted example is not used

Sent with the photo and the sentence, smart search answers: *formula ignored: 3 oxide(s), no flux - a formula needs at least 3 oxides including a flux (Na2O, K2O, Li2O, CaO, MgO, ZnO, BaO, SrO ...)*. The results are the same as 8b. A formula needs at least 3 oxides **including a flux** (CaO, K_2O, Na_2O, MgO, ZnO ...). Colour match never takes a formula: its chemistry is only on the recipe side.

10 · A colour word in the text replaces the photo's colour, so leave it out when you use a photo

On the page, "Read it" copies the text's colour into the colour box: "deep blu" → **#0000FF** replaces the measured **#0E1F3E**. The same search as 8b then gives:



#1 Floating Base + Copper + Chrome + Cobalt
Glazy #700509 · cone 5 - 6 · Glossy
cobalt blue #335470 · ΔE00 16.7
score **0.730** = text 0.22 · photo 0.20 · colour 0.10 · cone 0.08 · surface 0.08 · atmosphere 0.04



#2 Blueberry
Glazy #615101 · cone 5 1/2 - 8 · Glossy
navy blue #2D4087 · ΔE00 17.1
score **0.730** = text 0.24 · photo 0.19 · colour 0.10 · cone 0.08 · surface 0.08 · atmosphere 0.04



#3 VanGogh Blue
Glazy #636675 · cone 4 - 6 · Glossy
light gray #D5D6D0 · ΔE00 17.8
score **0.728** = text 0.23 · photo 0.19 · colour 0.10 · cone 0.08 · surface 0.08 · atmosphere 0.04

ΔE00 ≈ 17 from your bowl, and #3 is light grey: the colour part now compares with the wrong blue.

Written without a colour word: `glossy floating glaze, variegated with lighter streaks, cone 6 oxidation` → read as cone 6 · Glossy · Oxidation · effect mottled; the photo colour #0E1F3E is kept:



#1 Opal Blue
Glazy #794705 · cone 6 - 7 · Glossy
blue black #202538 · ΔE00 5.5
score **0.781** = text 0.19 · photo 0.21 · colour 0.18 · cone 0.08 · surface 0.08 · atmosphere 0.04



#2 Studio Neves Azul GA42
Glazy #97643 · cone 5 1/2 - 6 · Glossy
blue black #252B3B · ΔE00 3.0
score **0.780** = text 0.17 · photo 0.20 · colour 0.21 · cone 0.08 · surface 0.08 · atmosphere 0.04



#3 Cobalt Blue
Glazy #741292 · cone 5 - 7 · Glossy
navy blue #13183B · ΔE00 4.3
score **0.771** = text 0.18 · photo 0.19 · colour 0.19 · cone 0.08 · surface 0.08 · atmosphere 0.04

Without a photo, do put the colour in the words: then the text is the only source of colour.

Summary: what runs where

Step	Pretrained model / algorithm	Trained on our data	From the team
1 find and cut out	Grounding DINO base, SAM 2.1 large, OpenCLIP ViT-B/32 (zero-shot)	–	–
2 colours	ColorThief, k-means in CIELAB, CIEDE2000 (no model)	–	–
3 attributes	Grounding DINO (zero-shot evidence)	logistic regression	–
4 read the words	rules, difflib, Color Lab V3	TF-IDF + logistic regression	–
7a colour ranking	CIEDE2000 (no model)	–	colour score, surface and cone rules, catalog
7b colour match	all-MiniLM-L6-v2 (frozen)	–	text head, chemistry head, PCA-16, latent space, 4 / 4 / 1 / 1, K-Means clusters
8 smart search	all-MiniLM-L6-v2, DINOv2 ViT-L/14 (frozen)	text, photo and formula heads; K-Means families	training method (masked InfoNCE, composition groups), Explore rule · cleaned recipe tags